

Theory of Computation

Submission link: <https://forms.gle/9QfcWiqewGmCcvG36>

Exercise 1: (Mathematic preliminary, Language, String)

1. Let $\Sigma = \{a, b\}$ and $L = \{aa, bb\}$. Describe $\bar{L}$ by a set notation.

$$\begin{aligned}\bar{L} &= \Sigma^* - L \\ &= \{a, b\}^* - \{aa, bb\} \\ &= \{\lambda, a, b, ab, ba\} \cup \\ &\quad \{w \in \{a, b\}^* : |w| \geq 3\}\end{aligned}$$

2. Find five strings which are in each of the following languages.

a) $L = \{w \in \{a\}^*: |w| \bmod 3 \neq |w| \bmod 2\}$

$$\{aa, aaa, aaaa, aaaaa, aaaaaa\}$$

b) $L = \{w \in \{a, b\}^*: n_a(w) \geq n_b(w) + 1\}$

Where $n_a(w)$ means the number of a's in string w.

$$a, aa, aaa, aab, aba$$